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To the Editors,
Journal of Rail Transport Planning & Management

I submit the manuscript "Graph reinforcement learning versus NSGA-II for fairness-aware metro line extension at matched simulation budgets" for consideration as a research article.

The paper compares a weight-conditioned graph reinforcement learning policy with NSGA-II, scalarised genetic algorithms, tabular Q-learning, a greedy rule and random lines on the public Xi'an and Amsterdam metro expansion benchmarks. Each method evaluates about 10^4 simulated lines, and results are averaged over ten seeds. The paper corrects three weaknesses of the benchmark's usual scoring: the summed group percentage used as a demand figure, a Gini-based fairness term that rewards lines that serve nobody, and a uniform coverage proxy for accessibility. It also reports a leave-one-line-out test on Xi'an and a paired cross-city warm-start study.

The main result is negative for the learned policy. NSGA-II gives better deployable trade-off sets on both grids, and the learned policy returns nearly the same line for different weight vectors. I report this together with the analysis plan that fixed the endpoints before the evaluation runs. The work fits the journal's interest in network design and in quantitative and algorithmic methods for rail planning, and the findings should help planners choose between learned and evolutionary methods.

The manuscript is not under consideration elsewhere. All code, run records and the scripts that generate every table are available at `https://github.com/vinhqdang/metro_expansion`.

Sincerely,

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